

LOT COMPUTER

Build and Buying Manual

Product Spec · Bill of Materials · Roadmap

LOT Systems Corporation – Draft v0.1 – 2026.07.27

Status: PRE-PROTOTYPE – PLANNING PHASE

Compiled from:

docs/hardware/LOT-COMPUTER-PRODUCT-SPEC.md

docs/hardware/LOT-COMPUTER-BOM.md

docs/hardware/LOT-COMPUTER-ROADMAP.md

Firmware and software companion manuals are separate PDFs
in this same docs/hardware/manuals/ directory.

=====

LOT SYSTEMS CORPORATION
LOT COMPUTER – PHYSICAL PRODUCT SPECIFICATION
FIRST HARDWARE REALIZATION OF THE QIoT™ / LOT TERMINAL VISION

=====

DOCUMENT **PRODUCT-SPEC / LOT-COMPUTER v0.1 (DRAFT)**
ISSUE DATE 2026.07.27
AUTHOR Vadik Marmeladov (S-2), COSMO® CIA, LOT Systems
CLASS INTERNAL / PLANNING
STYLE TERMINAL GRID
STATUS PRE-PROTOTYPE – PLANNING PHASE

=====

00 WHAT THIS IS

LOT Computer is the first commercial physical device in the QIoT™ (Quantum Internet of Things) line referenced in the LOT Wiki as "LOT® Station / COSMO® node" and specified architecturally in `docs/corporate/LOT-TERMINAL-\*` as the S-2 operator hardware track. This document turns that vision into a buildable object: a coin-sized stainless steel terminal that sits on a desk, watches one weather metric, and speaks to its owner in one short line – "Coffee time!" – pushed live from lot-systems.com.

This spec does not replace `LOT-TERMINAL-VISION.md`, `LOT-TERMINAL-M2M.md`, or `LOT-TERMINAL-SYNC.md` – it is the first concrete SKU built on top of them. Where those documents describe the open-source S-2 operator platform and the M2M data-intake protocol in the abstract, this document specifies one exact, orderable, manufacturable object: ****LOT Computer – Node Zero****.

ONE OBJECT. ONE SCREEN LINE. ONE BUTTON. ONE SENSOR. ZERO CONFIGURATION.

Companion documents (this session):

- `LOT-COMPUTER-BOM.md` – components buying list, links, unit + 100-run cost
- `LOT-COMPUTER-ROADMAP.md` – phased plan from spec to 100-unit run
- `LOT-COMPUTER-FIRMWARE-SPEC.md` – on-device firmware document (separate, per brief pt.9/11)
- `LOT-COMPUTER-SOFTWARE-SPEC.md` – lot-systems.com-side software document (separate, pt.10/11)
- `manuals/LOT-Computer-Build-and-Buying-Manual.pdf` – PDF manual export (pt.7)

=====

01 THE 19-POINT BRIEF, MAPPED

The originating brief (Vadik, 2026-07-27) is reproduced below as 19 numbered logic items, each mapped to where it is specified in this document set. This mapping exists so no line item silently drops between planning and build.

```

| #  | BRIEF ITEM                                   | SPECIFIED IN                    |
|----|----------------------------------------------|---------------------------------|
| 1  | PCB Way                                      | §04 MANUFACTURING · BOM §02     |
| 2  | Pager-like notification from AI-powered site | §03 NOTIFICATION PROTOCOL       |
| 3  | 2-part stainless steel body                  | §02 INDUSTRIAL DESIGN           |
| 4  | Flat silver square 4x4cm x 5mm height        | §02 INDUSTRIAL DESIGN           |
| 5  | Camera                                       | §02 · FIRMWARE-SPEC §02         |
| 6  | Use LOT API connector                        | §03 · SOFTWARE-SPEC §02         |
| 7  | Result in PDF manuals                        | manuals/ · ROADMAP §05          |
| 8  | Compress the information in each session     | sessions/ (this practice)       |
| 9  | Firmware documents                           | FIRMWARE-SPEC.md (separate doc) |
| 10 | Software to connect with firmware            | SOFTWARE-SPEC.md (separate doc) |
| 11 | Separate documents                           | 6-document set, this session    |
| 12 | Charger                                      | §02 · BOM §02.6 (Qi wireless)   |
| 13 | 100 units run                                | ROADMAP §04 · BOM §04           |
| 14 | Weather sensor                               | §02 · BOM §02.5 (BME280-class)  |
| 15 | AI-grade off-the-shelf sensors               | BOM §02.5 (rationale)           |
| 16 | "Copy" button → signal to site Log tab       | §03 NOTIFICATION PROTOCOL       |
| 17 | One side polished stainless steel            | §02 INDUSTRIAL DESIGN           |
| 18 | Other side: camera, screen, button           | §02 INDUSTRIAL DESIGN           |
| 19 | Wireless charger                             | §02 · BOM §02.6                 |

```

=====

02 INDUSTRIAL DESIGN – THE OBJECT

FORM

A flat square tile, 40mm x 40mm x 5mm, machined in two stainless steel halves that clamshell around an internal PCB stack.

...

SIDE A
(FACE UP)
"the coin"

LOT

Mirror-polished 316 stainless.
Engraved or laser-etched
wordmark only. No screen,
no seams visible from top.

SIDE B
(FACE DOWN /
desk-facing
or wall-facing)

○

[====]
[SCRN]

(●)

Camera aperture (top-left) ·
round/square micro-display
(center-right) · single
mechanical/capacitive
"Copy" button (bottom-left)

...

- ****Side A – polished stainless steel**** (brief pt.17): the object at rest. Sits on a desk like a coin or a paperweight. No visible electronics. This is the face shown when the device is idle / no active notification.
- ****Side B – camera, screen, button**** (brief pt.18): the working face. Flipped up or wall-mounted at an angle so the screen is readable. Houses the camera aperture, the notification micro-display, and the Copy button.
- Two halves join with 4 countersunk M1.6 screws (or laser-welded seam for the production run once tolerances are proven) around a compressed silicone gasket – light dust/splash resistance, not a full IP rating.

THE 5MM CONSTRAINT – STATED HONESTLY

5mm total height is the aspirational industrial-design target stated in the brief. It is aggressive. Logged here so it is never silently dropped or silently "solved" by pretending it was easy:

...

STACK BUDGET AT 5MM (top shell + bottom shell + internal stack)

Stainless shell (top)	0.6mm	machined, not sheet
Stainless shell (bottom)	0.6mm	machined, not sheet
Gasket + clearance	0.3mm	
PCB (flex or rigid-flex)	0.4mm	
Camera module (bare, no lens stack barrel)	1.0–1.3mm	ultra-thin modules only
Display module	0.8–1.2mm	flexible OLED, not glass LCD
Battery OR supercap	1.0–1.5mm	thin-film LiPo, small capacity
Qi receiver coil (printed)	0.3mm	printed/flex coil, not off-shelf puck
TOTAL	≈5.0–5.8mm	TIGHT – near zero margin

...

CAUTION: this stack is achievable only with (a) a bare-die or chip-scale-package MCU, (b) a flexible-OLED or micro-LED display, not a glass-backed LCD, (c) a thin-film battery under 60mAh or a supercap bridging to the Qi coil, and (d) a lensless or pinhole camera rather than a standard M12 lens barrel. This is a valid V2/V3 miniaturization target. It is NOT the safe path for the first 100-unit run.

RECOMMENDATION: build the 100-unit pilot run at 40mm x 40mm x 9mm (still a flat coin, still two-part stainless, same face layout) to use off-the-shelf camera/display/battery modules with normal tolerances, then miniaturize to the true 5mm target for a v2 run once firmware and the notification/M2M protocol are proven on real desks. See ROADMAP §02.

SIDE B COMPONENT LAYOUT

Component	Role	See
Camera	Ambient capture, future gesture/presence sensing	§02.5, BOM §02.3
Micro-display	Renders one pager-style line pushed from the site	§03, BOM §02.4
"Copy" button	Press → writes an entry to the user's Log tab	§03, BOM §02.7
Weather sensor	Ambient temp / humidity / pressure, feeds AI grade	§02.5, BOM §02.5
Wireless charge coil	Qi inductive charging, no port, no cable to break	§02.6, BOM §02.6

LOT Computer – Build and Buying Manual – page 3

02.5 SENSORS – "AI-GRADE OFF-THE-SHELF"

Brief pt.14/15 calls for a weather sensor built from off-the-shelf parts rated for continuous, unattended, AI-consumed telemetry – not hobbyist-grade parts that drift. This maps directly onto the existing M2M "Multi-Sensor Array" format already specified in `docs/corporate/LOT-TERMINAL-M2M.md` (the "Psychotronic Weather Station" example): temperature, humidity, pressure, air quality. LOT Computer ships one combined environmental sensor (BME280-class: temp + humidity + pressure) as the v1 payload, with the M2M JSON shape already defined and unchanged – LOT Computer is simply the first *factory-built, single-SKU* S-2 device, where the existing docs describe hand-built maker devices.

02.6 CHARGER

Qi-standard wireless charging (brief pt.12/19) – no port on the shell, nothing to wear out, consistent with "no visible electronics on Side A." Device ships with a small stainless charging puck styled to match. See BOM §02.6 for the receiver IC and coil.

=====

03 NOTIFICATION PROTOCOL – THE PAGER LINE + THE COPY BUTTON

Brief pt.2 and pt.16 are one feature, split across two directions:

INBOUND – SITE → DEVICE ("Coffee time!")

```
...
lot-systems.com AI layer (Memory Engine / QIE, existing)
    | generates a short recommendation string
    | (same shape as the M2M "recommendation" field
    | already defined in LOT-TERMINAL-M2M.md, e.g.
    | "Open your windows for 3 minutes")
    |
    ▼
Notification queue (NEW – see SOFTWARE-SPEC §02)
    | one line, <=24 characters, plain text
    | e.g. "Coffee time!"
    |
    ▼
Device pairing channel (NEW – device_id + operator token,
    same auth shape as LOT-TERMINAL-SYNC.md Layer 1)
    |
    ▼
LOT Computer micro-display
    renders the line, wakes briefly, returns to idle
...

```

No existing push/webhook/SSE-to-device channel exists in the codebase today (confirmed by repository research this session – see SOFTWARE-SPEC §01). This is net-new server-side surface, deliberately built as a thin extension of the M2M protocol already on paper rather than a new one.

OUTBOUND – BUTTON → SITE LOG TAB

```
...
User presses the Copy button on Side B
    |
    ▼
Firmware timestamps the press, attaches last-shown notification text
    |
    ▼
POST to LOT API (device-authenticated, same channel as sensor telemetry)
    |
    ▼
Server writes one Log entry for that user
    (existing model: src/server/models/log.ts,
    existing endpoint family: GET/POST /api/logs,
    src/server/routes/api.ts:1082)
    |
    ▼
Entry appears in the user's Log tab exactly like a manual journal entry,
tagged with event metadata so the UI can render a device glyph
...

```

"Copy" names the action precisely: the button copies whatever the device is currently showing into the permanent record on the site, the same way a person would copy a line from a notepad into their journal.

=====

04 MANUFACTURING – PCBWAY

Brief pt.1. PCBWay is the specified fabricator for both the PCB and (via their CNC/sheet-metal service line) the two stainless steel shells, so a single vendor relationship covers electronics + enclosure for the pilot run.

```

'''

```

STAGE	PCBWAY SERVICE	QUANTITY (pilot)
PCB fab + assembly	PCB Prototype + PCBA	10 (bring-up) → 100 (run)
Stainless shells (x2)	CNC Machining service	10 (bring-up) → 200 (run, 2/unit)
Stencil (if hand-assembled)	SMT stencil	1

```

'''

```

Order sequence: 10-unit bring-up batch first (proves the stack in §02 and the firmware in FIRMWARE-SPEC.md), THEN the 100-unit run (brief pt.13). Never place the 100-unit order against an unverified board revision – see ROADMAP §03 for the explicit gate between bring-up and run.

05 ONE-LINE POSITIONING

LOT Computer is the desk-sized, silent half of the LOT loop: the software already listens, predicts, and remembers everything a user does – LOT Computer is the object that occasionally interrupts, in one line, with the one thing worth saying out loud, and the one button that lets the user talk back without opening a laptop.

```

=====
LOT SYSTEMS CORPORATION                                LOS ANGELES, CA
END OF SPECIFICATION – DRAFT v0.1                      2026.07.27
=====

```

```

=====
LOT SYSTEMS CORPORATION
LOT COMPUTER – BILL OF MATERIALS / BUYING LIST
=====

```

```

DOCUMENT      BOM / LOT-COMPUTER v0.1 (DRAFT)
ISSUE DATE    2026.07.27
PARENT        LOT-COMPUTER-PRODUCT-SPEC.md §02, §04
STYLE         TERMINAL GRID
NOTE          Part numbers below are real, commercially available parts chosen
              for realistic sourcing and costing. Exact SKUs/links must be
              re-verified against live distributor stock before the bring-up
              order – component availability and pricing shift weekly.

```

00 HOW TO READ THIS LIST

Two build targets, per PRODUCT-SPEC §02's honest sizing note:

```

V1 PILOT      – 40 x 40 x 9mm shell, off-the-shelf modules, LOW RISK
V2 TARGET     – 40 x 40 x 5mm shell, bare-die/flex modules, HIGH RISK

```

This BOM prices V1 PILOT (the buildable one) and flags the V2 substitution where it differs. Order the V1 pilot first – see ROADMAP §02/§03.

01 MANUFACTURING VENDOR

Item	Vendor	Link
PCB fab + PCBA (assembly)	PCBWay	https://www.pcbway.com
CNC stainless shells (x2)	PCBWay CNC service	https://www.pcbway.com/rapid-prototyping/CNC_Machining.html
SMT stencil	PCBWay	https://www.pcbway.com

PCBWay is specified per brief pt.1 for single-vendor accountability across electronics + enclosure on the pilot run.

02 COMPONENTS

02.1 COMPUTE / RADIO – MCU

Part	Role	Source	Link
Espressif ESP32-S3-WROOM-1	MCU + WiFi + BLE, camera-capable (DVP)	Espressif / distributors	https://www.espressif.com/en/products/modules
Digi-Key / Mouser (distribution)	Distributor for ESP32-S3 + passives	Digi-Key, Mouser	https://www.digikey.com · https://www.mouser.com

ESP32-S3 chosen over a simpler MCU specifically because it has a native camera (DVP) interface and enough WiFi + crypto headroom to hold the device-auth token for the LOT API connector (SOFTWARE-SPEC §02) without a companion chip. V2 target: ESP32-S3-MINI-1 (smaller footprint, same silicon) or a chip-scale variant once the 5mm shell is attempted.

02.2 BATTERY

Part	Role	Source	Link
3.7V LiPo, 100–150mAh, thin (V1)	Power buffer between Qi charge cycles	Adafruit / Digi-Key	https://www.adafruit.com
3.7V thin-film LiPo <60mAh (V2)	Same, sized for 5mm stack gate	Specialty cell vendors (TBD at V2)	–

02.3 CAMERA

Part	Role	Source	Link
OV2640 camera module (DVP, 2MP)	Ambient capture / future presence sensing	Adafruit, generic DVP-camera modules	https://www.adafruit.com
V2: ultra-thin lensless/pinhole camera module	5mm-stack substitute	TBD, evaluate at V2 gate	–

02.4 DISPLAY

Part	Role	Source	Link
0.96"–1.3" round/square OLED, SPI/I2C (SSD1306/GC9A01 class)	Pager-line micro-display	Adafruit, Waveshare	https://www.adafruit.com · https://www.waveshare.com
V2: flexible OLED module	5mm-stack substitute	TBD, evaluate at V2 gate	–

02.5 SENSORS – "AI-GRADE OFF-THE-SHELF"

Part	Role	Source	Link
Bosch BME280	Temperature + humidity + pressure, I2C	Adafruit, Digi-Key, Mouser	https://www.adafruit.com · https://www.bosch-sensortec.com
(Optional v1.1) Sensirion SGP40	Air-quality (VOC) index, extends M2M sensor array to match LOT-TERMINAL-M2M.md example	Digi-Key, Mouser	https://www.sensirion.com

BME280 is chosen specifically because it is the same class of part already named as the reference sensor payload in `docs/corporate/LOT-TERMINAL-M2M.md` ("Temperature", "Humidity", "Barometric Pressure" fields) – the device ships speaking the protocol that already exists on paper, no new data shape required for v1.

02.6 WIRELESS CHARGING

Part	Role	Source	Link
TI BQ51013B (Qi receiver IC)	Inductive charge receiver, regulates to battery	Digi-Key, Mouser, TI direct	https://www.ti.com/product/BQ51013B
Receiver coil (Qi, thin profile)	Paired inductive coil	Digi-Key, Würth Elektronik	https://www.wurth-electronic.com
Matching Qi charging puck (stainless-styled, custom shell)	User-facing charger accessory	PCBWay CN	
C + generic Qi transmitter module			https://www.pcbway.com

02.7 INPUT

Part	Role	Source	Link
Low-profile tactile switch OR capacitive touch pad (CAP1203-class)	"Copy" button	Digi-Key, Mouser	

02.8 ENCLOSURE

Part	Role	Source	Link
316 stainless steel, CNC 2-piece shell, one face mirror-polished	Body (brief pt.3, 17, 18)	PCBWay	https://www.pcbway.com/rapid-prototyping/CNC_Machining.html
Silicone gasket, custom-cut	Dust/splash seal between halves	PCBWay or local gasket supplier	https://www.pcbway.com
M1.6 countersunk screws (x4)	Shell fastening (bring-up only; production = laser weld)	McMaste	https://www.mcmaster.com

03 UNIT COST – V1 PILOT (SINGLE UNIT, PROTOTYPE PRICING)

CATEGORY	PART	EST. UNIT COST (proto qty)
Compute/Radio	ESP32-S3-WROOM-1	\$3.50 – \$5.00
Battery	100–150mAh thin LiPo	\$2.50 – \$4.00
Camera	OV2640 module	\$4.00 – \$7.00
Display	0.96"–1.3" OLED module	\$4.00 – \$8.00
Sensors	BME280 (+ optional SGP40)	\$6.00 – \$12.00
Wireless charging	BQ51013B + coil	\$3.00 – \$5.00
Input	Button/touch part	\$0.50 – \$1.50
PCB fab + PCBA	PCBWay, proto qty	\$15.00 – \$25.00
Enclosure (2 shells)	PCBWay CNC stainless, proto qty	\$25.00 – \$45.00
Gasket + fasteners	misc	\$1.00 – \$2.00
TOTAL (proto, per unit)		≈ \$65 – \$115

Prototype-quantity pricing is intentionally pessimistic – the bring-up batch (10 units, ROADMAP §02) is expected to land at the top of this range or above due to PCBWay/CNC minimum-order pricing.

04 100-UNIT RUN COSTING (BRIEF PT.13)

CATEGORY	UNIT COST @ 100qty	x100 TOTAL
Electronics BOM (all)	\$18 – \$28	\$1,800 – \$2,800
PCB fab + PCBA	\$6 – \$10	\$600 – \$1,000
Enclosure (2 shells, CNC)	\$10 – \$18	\$1,000 – \$1,800
Gasket + fasteners	\$0.60 – \$1.20	\$60 – \$120
Qi charging puck accessory	\$6 – \$10	\$600 – \$1,000
SUBTOTAL, HARDWARE		≈ \$4,060 – \$6,720
Firmware flash + QC test jig (one-time tooling)		\$500 – \$1,200
Packaging (100 units)		\$300 – \$600
TOTAL, 100-UNIT RUN		≈ \$4,860 – \$8,520
PER-UNIT, FULLY LOADED		≈ \$49 – \$85

Component pricing drops meaningfully at 100qty vs. proto qty (roughly 2–3x reduction on the semiconductor line items) – this is the standard justification for the bring-up → run gate in ROADMAP §03: never place the 100-unit order until the bring-up batch has proven the design, because a respin at 100qty costs the whole run twice.

LOT SYSTEMS CORPORATION
END OF BOM – DRAFT v0.1

LOS ANGELES, CA
2026.07.27

DOCUMENT ROADMAP / LOT-COMPUTER v0.1 (DRAFT)
ISSUE DATE 2026.07.27
PARENT LOT-COMPUTER-PRODUCT-SPEC.md
STYLE TERMINAL GRID

00 SEQUENCE – BUILD ORDER

00 THIS SESSION – plan, BOM, roadmap, firmware spec, software spec (docs only)
01 Server-side scaffolding – device pairing + notification queue + M2M extension for the "Copy" write-back (SOFTWARE-SPEC §02). Software before hardware exists so the bring-up units have something real to talk to.
02 Bring-up batch – 10 units, V1 PILOT shell (40x40x9mm), off-the-shelf modules per BOM §02. Hand-assembled or PCBWay low-qty PCBA.
03 Bring-up gate – firmware + protocol proven on real desks before the 100-unit order is placed. See §03 below. NEVER SKIP THIS GATE.
04 100-unit production run – PCBWay PCBA + CNC shells at run pricing (BOM §04).
05 PDF manuals finalized from the by-then-stable spec/firmware/software docs (brief pt.7) and shipped in the box.
06 V2 miniaturization track (5mm shell) begins in parallel with V1 field use, not before – see §04 below.

Order matters, same discipline as `docs/technical/LOT-NODE-0-RIG-SPEC.md`
§05: the channel the device reports into must exist and be tested before the first unit leaves the bench, so there is never a batch of hardware with nothing real to talk to.

01 PHASE 0 – PLANNING (THIS SESSION)

Deliverable	Status
LOT-COMPUTER-PRODUCT-SPEC.md	DONE – this session
LOT-COMPUTER-BOM.md	DONE – this session
LOT-COMPUTER-ROADMAP.md (this file)	DONE – this session
LOT-COMPUTER-FIRMWARE-SPEC.md	DONE – this session
LOT-COMPUTER-SOFTWARE-SPEC.md	DONE – this session
PDF manual export	DONE – this session
Session report (compressed)	DONE – this session

No components ordered, no PCB fabricated, no code written against production in this phase – planning only, per the scheduled-task scope. Phase 1 is the first phase that touches the live LOT API surface.

02 PHASE 1 – SERVER-SIDE SCAFFOLDING

Task	Owns
Device pairing endpoint (device_id + operator token)	SOFTWARE-SPEC §02.1
Notification queue table + push-to-device endpoint	SOFTWARE-SPEC §02.2
"Copy" write-back endpoint → Log model	SOFTWARE-SPEC §02.3
M2M sensor-intake extension (reuses existing M2M shape)	SOFTWARE-SPEC §02.4
Admin test harness (simulate a device with curl/Postman)	SOFTWARE-SPEC §03

Exit criterion: a curl script can (a) register a fake device, (b) push "Coffee time!" to it and read it back, (c) POST a fake Copy-button event and see a Log entry appear in a test account – all before any physical unit exists.

03 PHASE 2 – BRING-UP BATCH (10 UNITS) → GATE → 100-UNIT RUN

Bring-up (10 units)

```
Order PCB (PCBWay proto), CNC shells x20 (2/unit), components per BOM §03
Hand-populate or low-qty PCBA
Flash FIRMWARE-SPEC.md v1 firmware
Pair each unit against the Phase-1 server scaffolding
Run on a real desk for >=7 consecutive days per unit
```

Gate – DO NOT SKIP

```
[ ] Qi charge cycle survives >=200 cycles on a sample unit (accelerated test)
[ ] Notification round-trip (site → screen) latency <5s, measured, 20 pushes
[ ] "Copy" button write-back appears in Log tab within 5s, 20/20 presses
[ ] BME280 telemetry matches a reference thermometer/hygrometer within spec
[ ] Battery lasts >=48h between Qi charges under normal notification cadence
[ ] Enclosure gasket passes a light-splash test, no ingress
[ ] Zero units in bring-up batch bricked by an OTA update test
```

Only once every box above is checked does the 100-unit PCBWay order go out.
A failed gate is not a delay – it is the gate doing its job.

100-unit run

```
Place PCBWay PCBA + CNC order at BOM §04 run pricing
QC jig: every unit flashed, paired, and given one live notification +
        one Copy-button test before packaging
Package with printed quick-start card + link to full PDF manual
```

04 PHASE 3 – V2 MINIATURIZATION (5MM TARGET)

Begins only after Phase 2's field data exists (real battery life, real thermal behavior, real firmware stability). Substitutes the parts flagged "V2" in BOM §02 (bare-die MCU package, flexible OLED, thin-film battery, lensless camera). Treated as a separate hardware revision, not a silent swap – a V1 unit and a V2 unit must both report `hw\_rev` in their M2M payload so server-side analytics can tell them apart.

05 PDF MANUALS (BRIEF PT.7) – STANDING PRACTICE

Each phase above regenerates the PDF manual set from the then-current markdown specs, not written fresh by hand:

Manual	Source markdown
Build & Buying Manual (this session)	PRODUCT-SPEC.md + BOM.md + ROADMAP.md
Firmware Manual (Phase 1+)	FIRMWARE-SPEC.md
Software / API Manual (Phase 1+)	SOFTWARE-SPEC.md
User Quick-Start Card (Phase 2+)	New, once UX is field-tested

Keeping the PDFs derived from markdown (not maintained separately) is the same "single source of truth" discipline the rest of this repo already uses for wiki/field-manual generation – see `docs/wiki/` and `docs/technical/LOT-FIELD-MANUAL-v53-SNAPSHOT.md` for the existing precedent.

LOT SYSTEMS CORPORATION
END OF ROADMAP – DRAFT v0.1

LOS ANGELES, CA
2026.07.27